SASTRA

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THINK MERIT | THINK TRANSPARENCY | THINK SASTRA

LABORATORY-SCALE PLANETARY SURFACE MAPPING USING USRP-BASED SYNTHETIC APERTURE RADAR

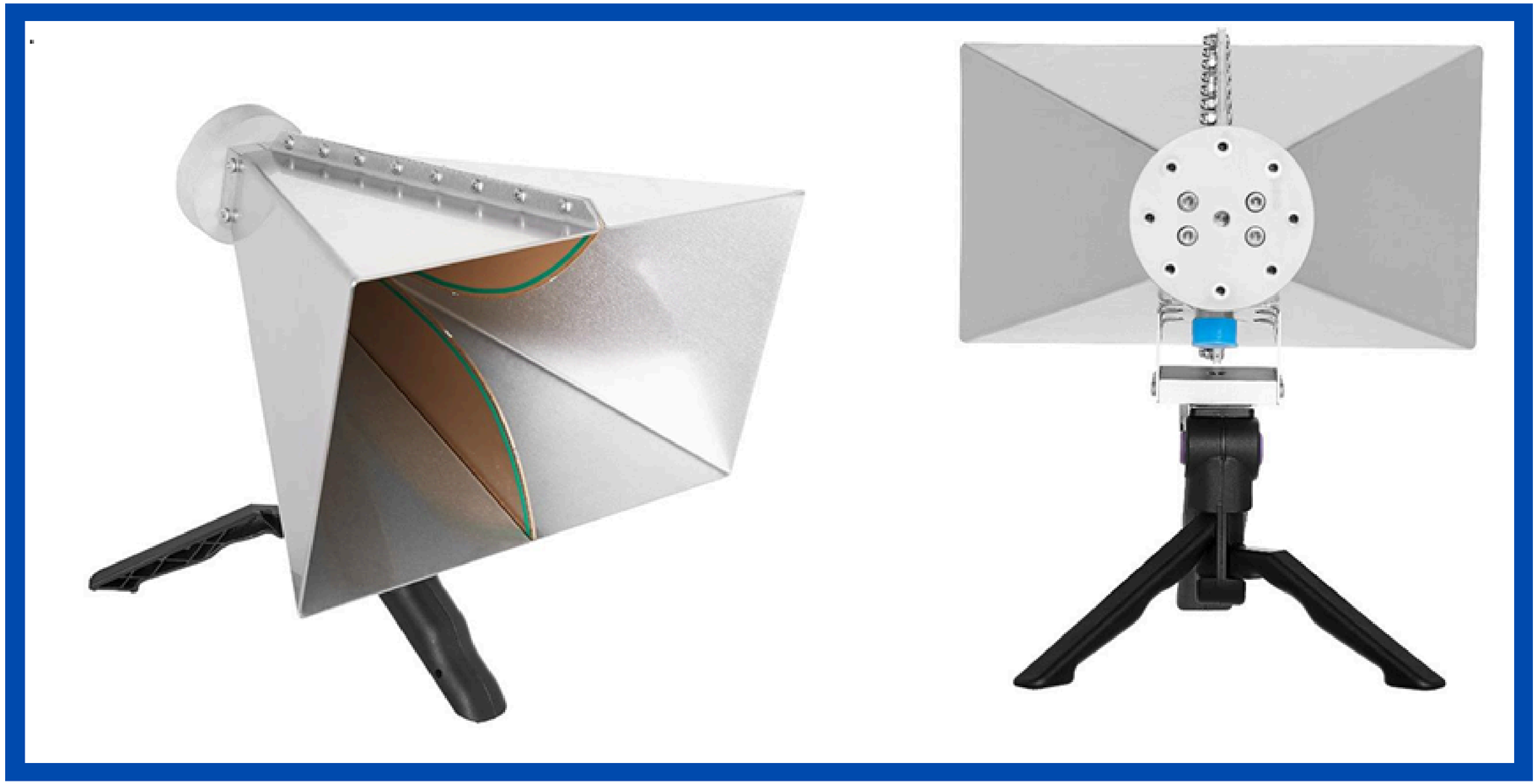
GUIDED BY:

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Professor
SEEE
SASTRA Deemed University

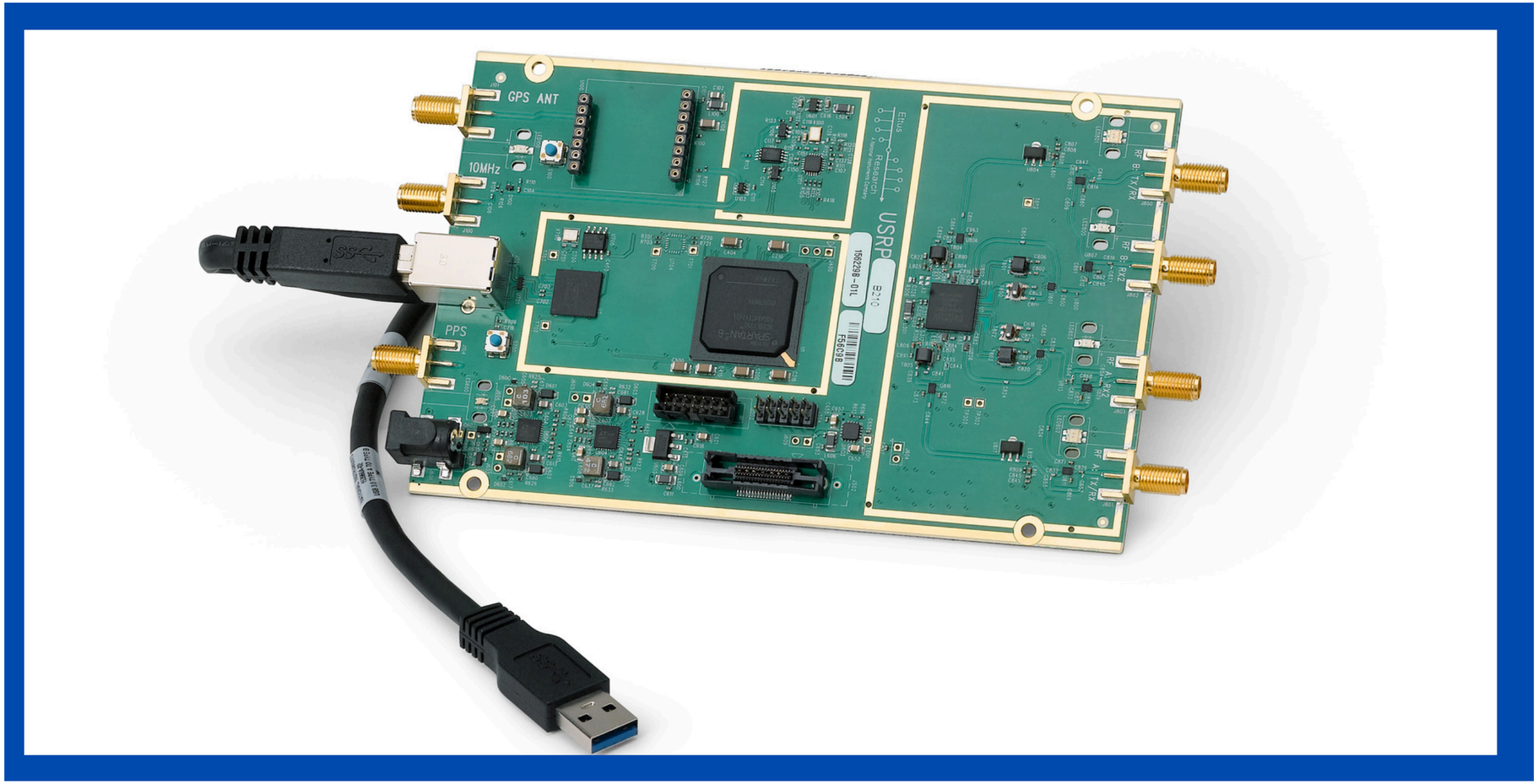
PRESENTED BY:

SUDEEPAN V (127004266)
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B.Tech. ECE

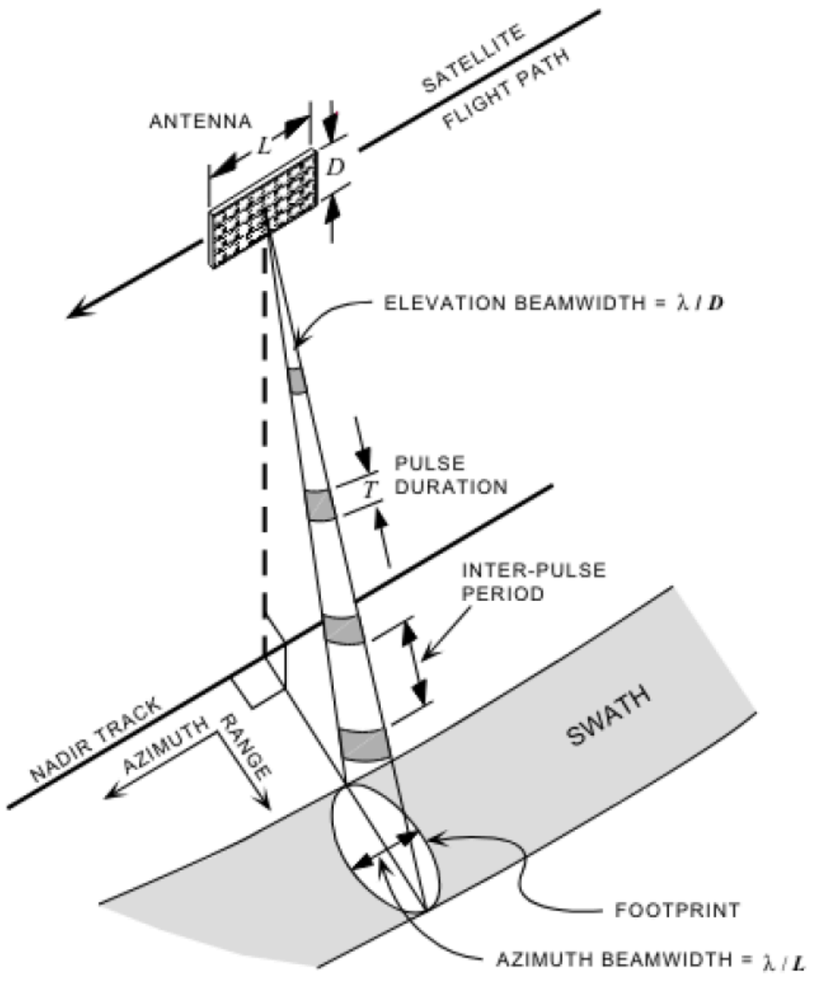
1 GHz – 8 GHz Double Ridged Horn Antenna



USRP B210

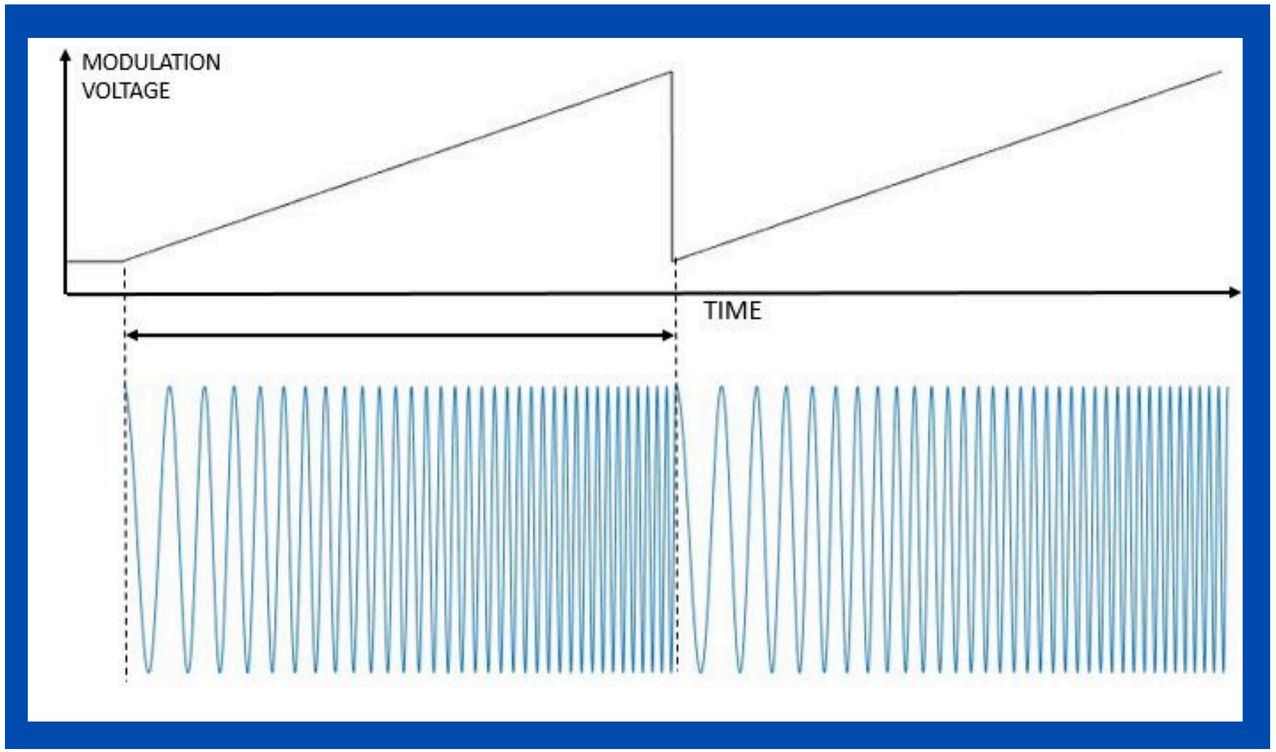


SPECIFICATIONS:



Parameters	Values
Operating Frequency	1 to 6 GHz
Effective Aperture	$0.0209\,m^2$
Physical Aperture	$0.037\,m^2$
Bandwidth	500 MHz
Azimuth Beamwidth	43.5°
Elevation Beamwidth	75.5°
Antenna height from ground	2 m
Ground Swath Length	3.09 m
Ground Swath Width	1.6 m
Azimuth Resolution	0.076 cm
Range Resolution	0.3 m

FMCW SAR



Transmit Chirp Signal

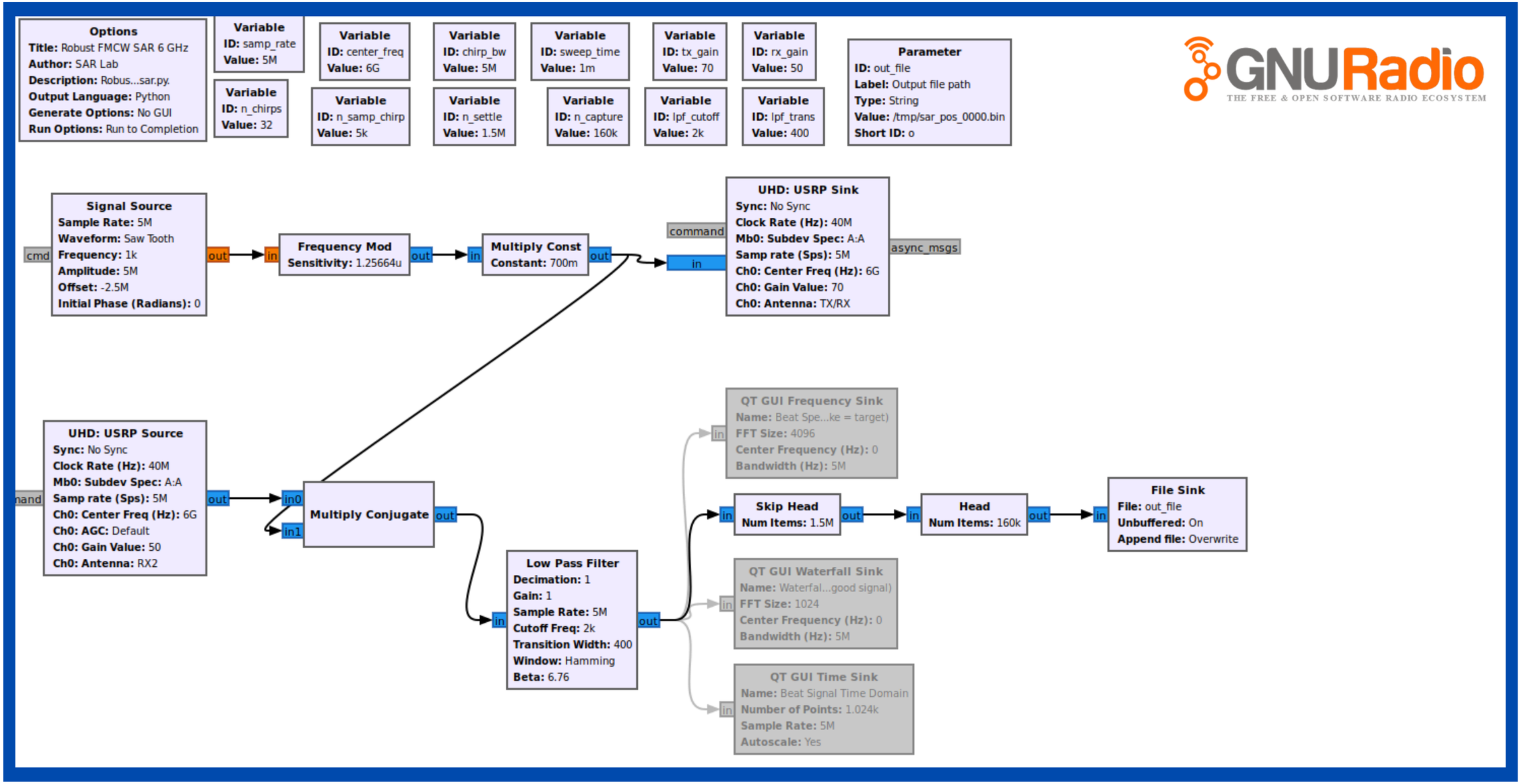
Receive Reflected Signal

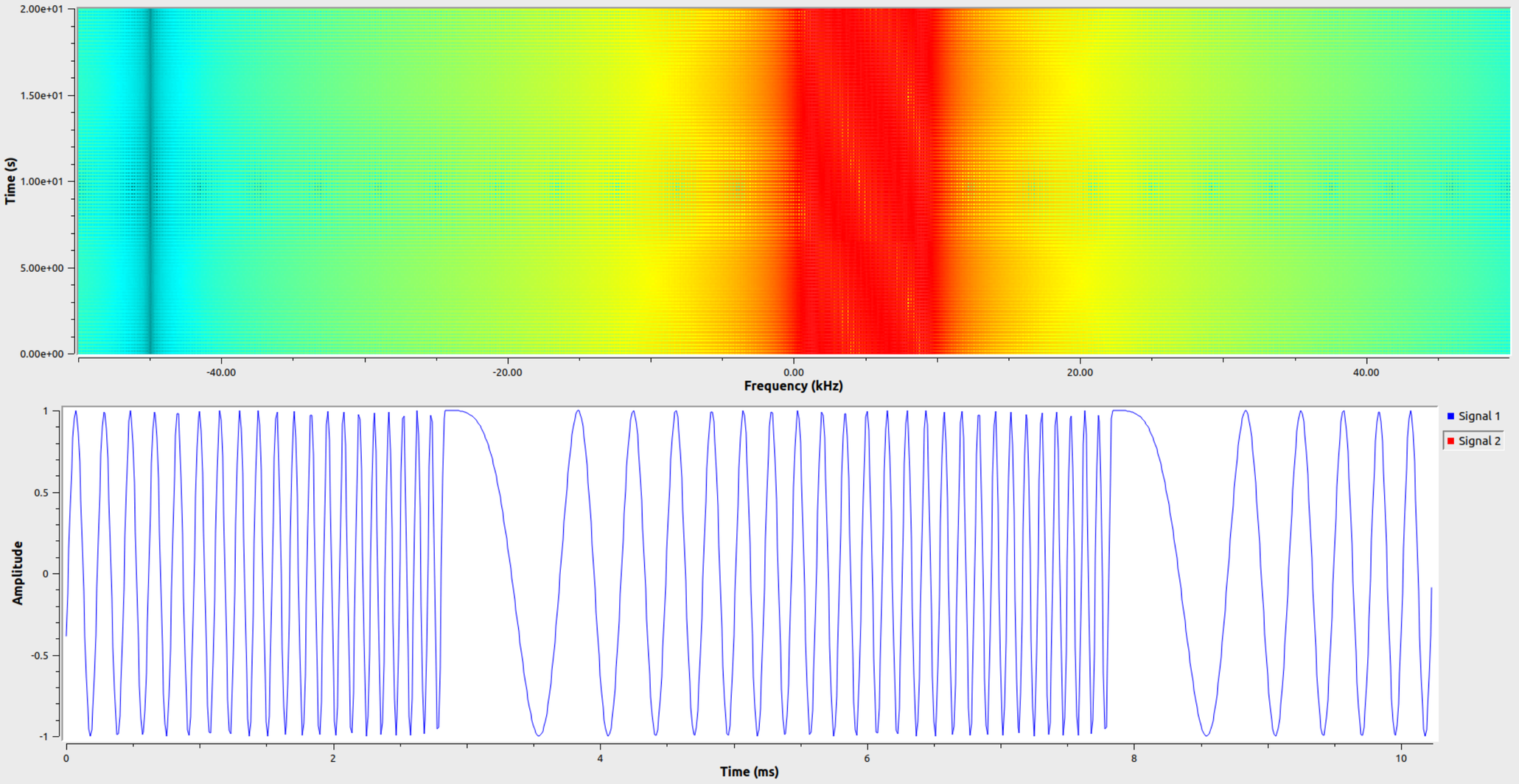
Mix : Generate Beat Signal

Repeat Across Position

Extract Range Information

Combine data to get SAR image


Options
Title: Robust FMCW SAR 6 GHz
Author: SAR Lab
Description: Robus...sar.py.
Output Language: Python
Generate Options: No GUI
Run Options: Run to Completion
Variable
ID: samp_rate
Value: 5M
Variable
ID: n_chirps
Value: 32
Variable
ID: center_freq
Value: 6G
Variable
ID: n_samp_chirp
Value: 5k
Variable
ID: chirp_bw
Value: 5M
Variable
ID: n_settle
Value: 1.5M
Variable
ID: sweep_time
Value: 1m
Variable
ID: n_capture
Value: 160k
Variable
ID: tx_gain
Value: 70
Variable
ID: lpf_cutoff
Value: 2k
Variable
ID: rx_gain
Value: 50
Variable
ID: lpf_trans
Value: 400
Parameter
ID: out_file
Label: Output file path
Type: String
Value: /tmp/sar_pos_0000.bin
Short ID: o
GNURadio
THE FREE & OPEN SOFTWARE RADIO ECOSYSTEM
Signal Source
Sample Rate: 5M
Waveform: Saw Tooth
Frequency: 1k
Amplitude: 5M
Offset: -2.5M
Initial Phase (Radians): 0
cmd
out
in
Frequency Mod
Sensitivity: 1.25664u
out
in
Multiply Const
Constant: 700m
out
command
in
UHD: USRP Sink
Sync: No Sync
Clock Rate (Hz): 40M
Mb0: Subdev Spec: A:A
Samp rate (Sps): 5M
Ch0: Center Freq (Hz): 6G
Ch0: Gain Value: 70
Ch0: Antenna: TX/RX
async_msgs
UHD: USRP Source
Sync: No Sync
Clock Rate (Hz): 40M
Mb0: Subdev Spec: A:A
Samp rate (Sps): 5M
Ch0: Center Freq (Hz): 6G
Ch0: AGC: Default
Ch0: Gain Value: 50
Ch0: Antenna: RX2
out
in0
in1
Multiply Conjugate
out
Low Pass Filter
Decimation: 1
Gain: 1
Sample Rate: 5M
Cutoff Freq: 2k
Transition Width: 400
Window: Hamming
Beta: 6.76
in
out
QT GUI Frequency Sink
Name: Beat Spe...ke = target)
FFT Size: 4096
Center Frequency (Hz): 0
Bandwidth (Hz): 5M
Skip Head
Num Items: 1.5M
Head
Num Items: 160k
File Sink
File: out_file
Unbuffered: On
Append file: Overwrite
QT GUI Waterfall Sink
Name: Waterfal...good signal)
FFT Size: 1024
Center Frequency (Hz): 0
Bandwidth (Hz): 5M
QT GUI Time Sink
Name: Beat Signal Time Domain
Number of Points: 1.024k
Sample Rate: 5M
Autoscale: Yes


2.00e+01
1.50e+01
1.00e+01
5.00e+00
0.00e+00
Time (s)
-40.00
-20.00
0.00
20.00
40.00
Frequency (kHz)
1
0.5
0
-0.5
-1
Amplitude
0
2
4
6
8
10
Time (ms)
Signal 1
Signal 2

System Parameters

Parameters	Values
Center frequency	6 GHz
Bandwidth	5 MHz
Sample rate	5 MSPS
Sweep Time	1 ms
Positions	64
Step Size	1.5 cm
Target Distance	~2.5 m



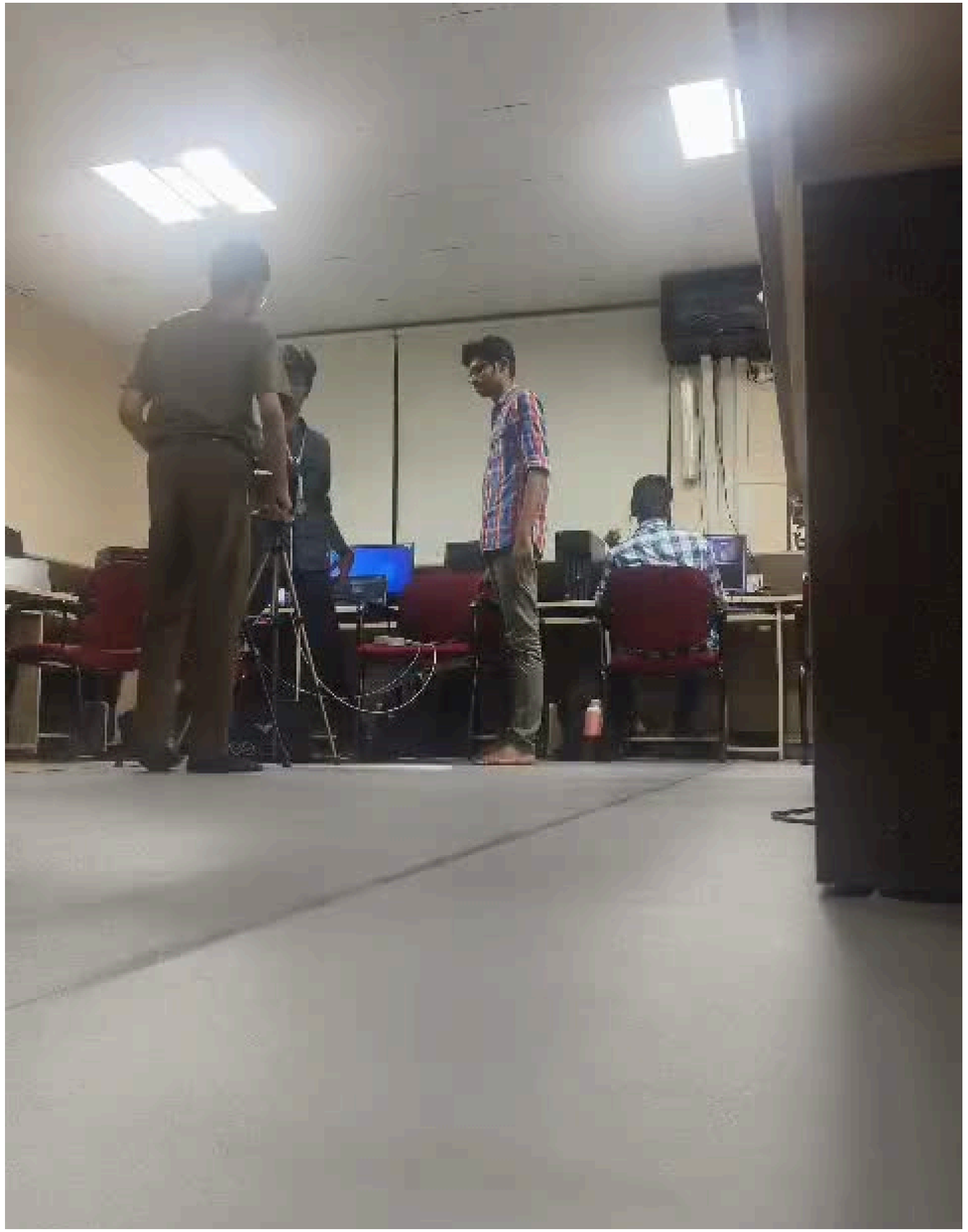
Data Acquisition Setup

```
girinath@groundstation: ~
[INFO] [B200] Operating over USB 3.
[INFO] [B200] Initialize CODEC control...
[INFO] [B200] Initialize Radio control...
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Performing register loopback test...
[INFO] [B200] Register loopback test passed
[INFO] [B200] Asking for clock rate 40.000000 MHz...
[INFO] [B200] Actually got clock rate 40.000000 MHz.
UUUUUUUUUusrp_source :error: In the last 1053 ms, 1 overflows occurred.
OUUUUUUUUUUUUUUUUUUUUUOUUUUUUUUUUUUUUUUUUUUUOUusrp_sink :error: In the last 752 ms, 49 underflows oc
curred.
UUUUUUUUUUUUUUUUUusrp_source :error: In the last 758 ms, 3 overflows occurred.
OUUUUUUUUUUUUUUUUUUUUUOUUUUUUUUUUUUUUUUUUUUUOUUUusrp_sink :error: In the last 758 ms, 57 underflows o
ccurred.
UUUUUUUUUUUUUUUUUUOUUUUUUUUUUUUUUUUUUUUUusrp_source :error: In the last 999 ms, 4 overflows occurred
.
OUUUUUUUUUUUUUUUUUUUUOUUUusrp_sink :error: In the last 760 ms, 57 underflows occurred.
UUUUUUUUUUUUUUUUUOUUUUUUUUUUUUUUUUUUUUUusrp_source :error: In the last 760 ms, 3 overflows occurred
.
OUUUUUUUUUUUUUUUUUUUUOUUUusrp_sink :error: In the last 759 ms, 57 underflows occurred.
UUUUUUUUUUUUUUUUOUUUUUUUUUUUUUUUUUUUUUOUUUUUUUUUUUUUUUUUUUusrp_source :error: In the last 999 ms, 4
 overflows occurred.
OUUUUusrp_sink :error: In the last 760 ms, 57 underflows occurred.
UUUUUUUUUUUUUUUOUUUUUUUUUUUUUUUUUUUUUOUUUUUUUUUUUUUUUUUUUOUUUUusrp_sink :error: In the last 760 ms
, 57 underflows occurred.
UUUUUUUUUUUUUusrp_source :error: In the last 987 ms, 4 overflows occurred.
OUUUUUUUUUUUUUUUUUUUUUOUUUUUUUUUUUUUUUUUUUUUOUUUUUusrp_sink :error: In the last 759 ms, 57 underflo
ws occurred.
UUUUUUUUUUUUusrp_source :error: In the last 759 ms, 3 overflows occurred.
OUUUUUU[00:08:26]   Captured 160,000 samples (1250 KB) in 8.5s
[00:08:26]   Progress: 18/64  ETA: 37.3 min

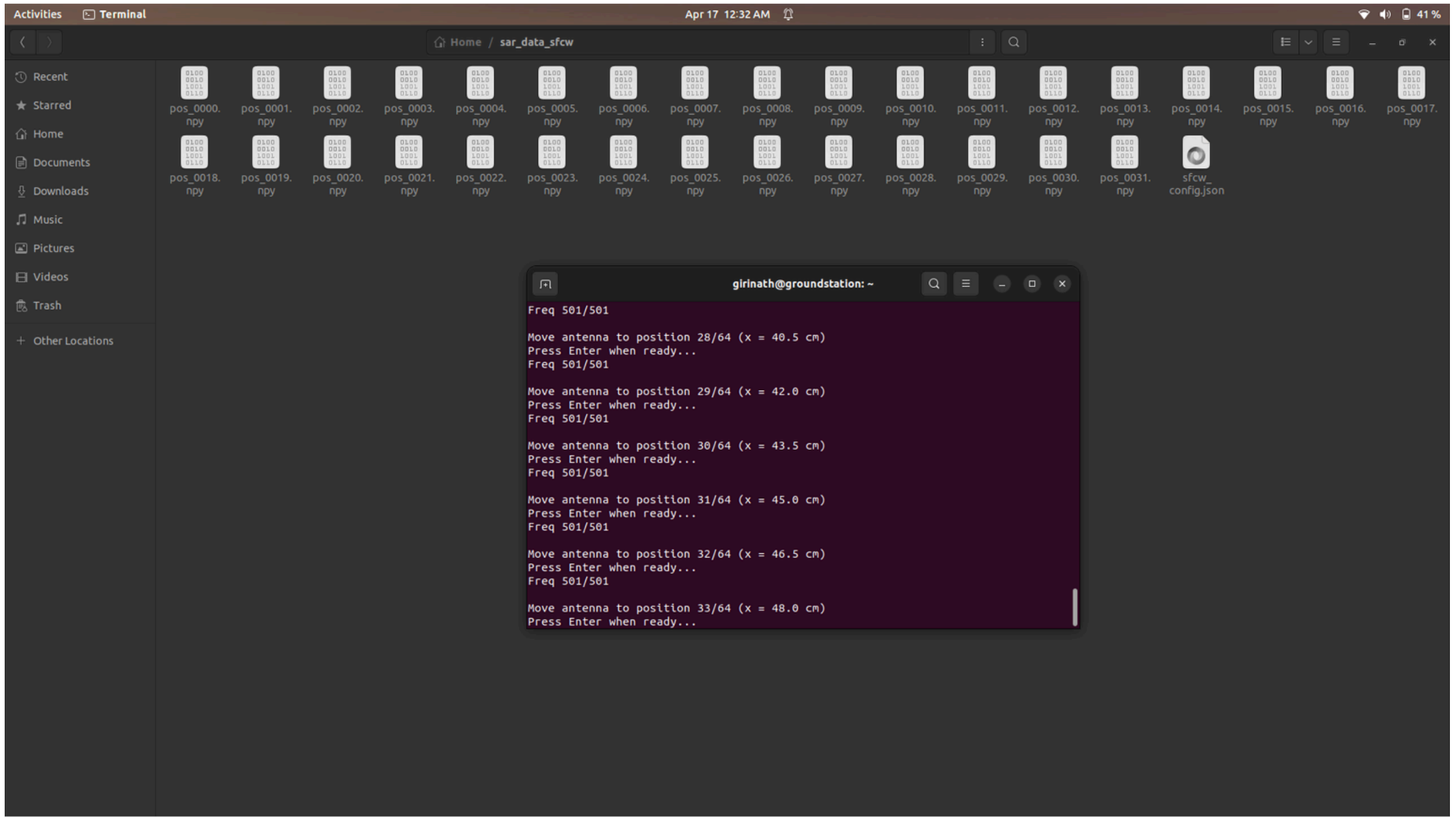
──────────────────────────────────────────────

[00:08:26] Position 19/64  x = 27.0 cm

  Move antenna from x = 25.5 cm  →  x = 27.0 cm
  (advance 15 mm = 0.1 cm)
  Press Enter when in position...
```



Data Structure



Captured Data:

IQ samples per position

Final Output:

SAR Matrix:

$S(x,t)$

Stored as:

- **.mat** file (matrix)
- **.json** config file

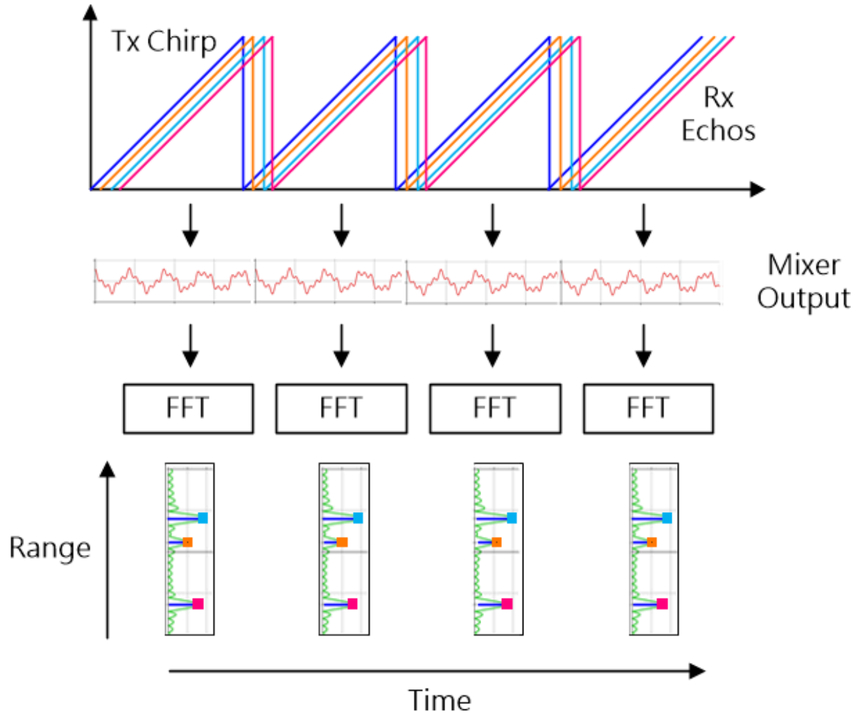
Matrix Structure

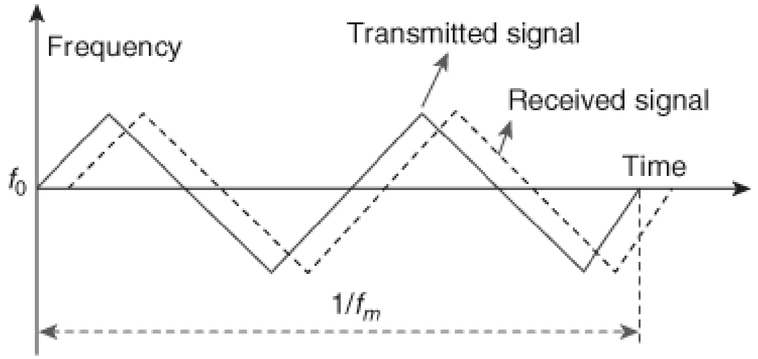
	t1	t2	t3	t4	t5	.	.	.	.	.	.	t5000
X1												
X2												
.												
.												
.												
X64												

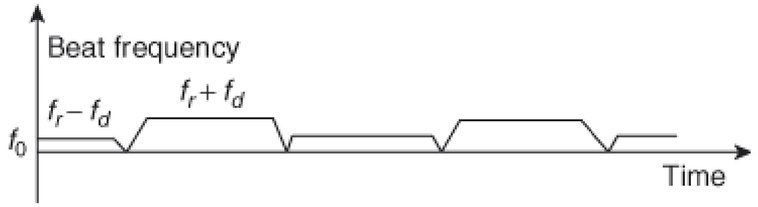
Range Calculation

$$S(x,t) \xrightarrow{\text{FFT}} S(x,f_b)$$

$$R = \frac{c.\,f_b}{2K}$$







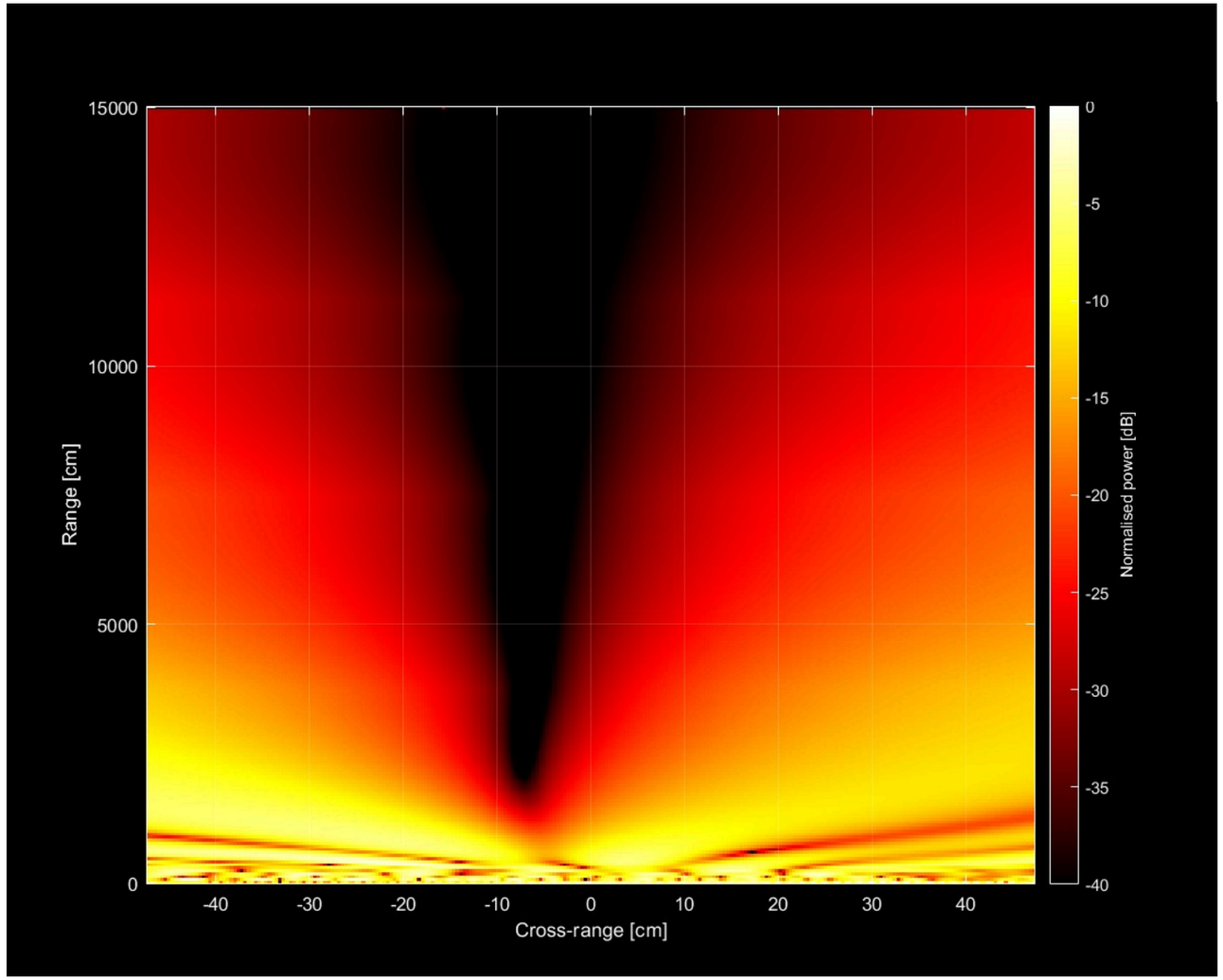
Signal Processing & Output

Processing Steps (MATLAB):

- **Background subtraction**
- **Windowing**
- **IFFT** → Range profiles
- **Back projection** → Image formation

Output: 2D SAR image (Range vs Cross-range)

RESULTS



Issues observed:

- Lower Bandwidth
- Unable to sample at higher rate (20 MSPS) which causes underflow and overflow
- Very low samples per chirp leads to poor IFFT resolution

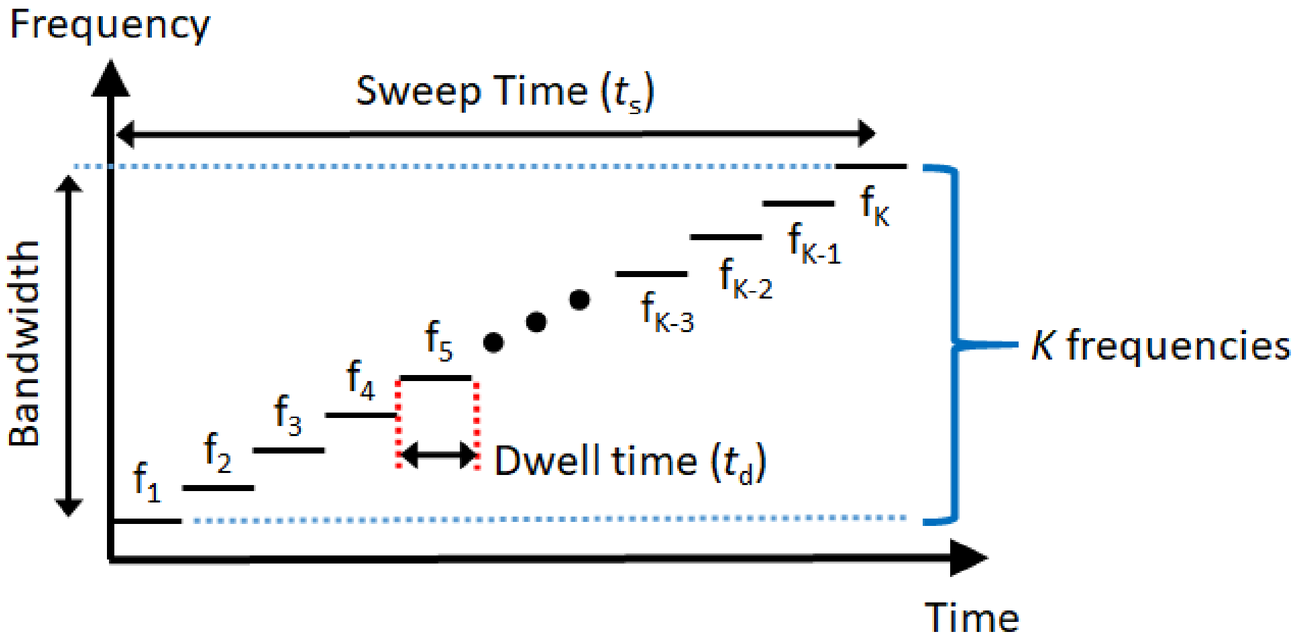
Conclusion:

FMCW implementation was not reliable for precise SAR imaging using USRP B210

Transition to SFCW Radar

New Approach:

Switched to Stepped Frequency Continuous Wave (SFCW)



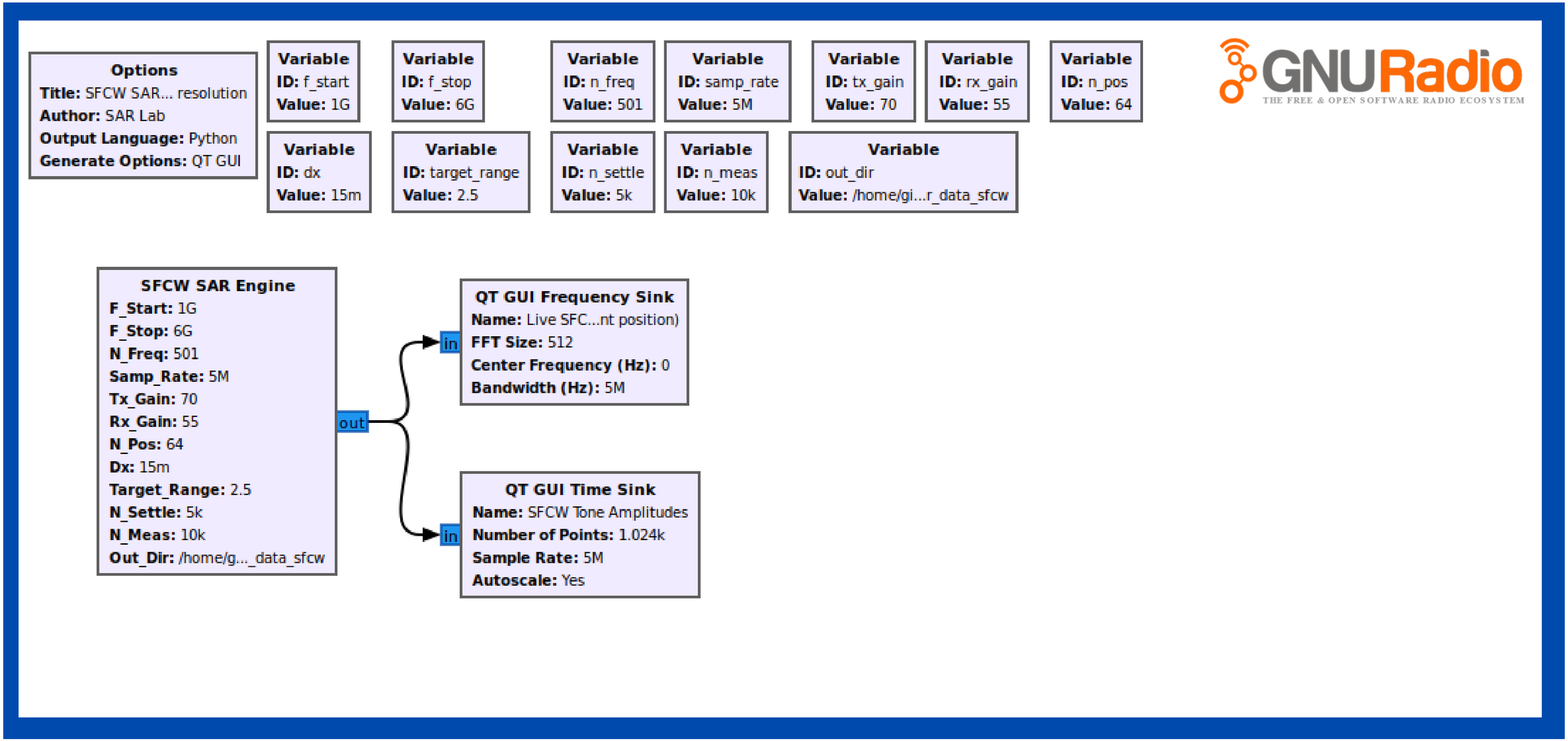
Reason for choosing SFCW:

- Easier to implement with SDR
- No need for continuous chirp generation
- Better control over frequency sweep
- Improved stability

SFCW System Parameters

Parameters	Values
Start Frequency	1 GHz
Stop Frequency	6 GHz
No. of Steps	501
Total Bandwidth	5 GHz
Sample Rate	5 MSPS
Positions	64
Step Size	1.5 cm




Options
Title: SFCW SAR... resolution
Author: SAR Lab
Output Language: Python
Generate Options: QT GUI
Variable
ID: f_start
Value: 1G
Variable
ID: f_stop
Value: 6G
Variable
ID: n_freq
Value: 501
Variable
ID: samp_rate
Value: 5M
Variable
ID: tx_gain
Value: 70
Variable
ID: rx_gain
Value: 55
Variable
ID: n_pos
Value: 64
Variable
ID: dx
Value: 15m
Variable
ID: target_range
Value: 2.5
Variable
ID: n_settle
Value: 5k
Variable
ID: n_meas
Value: 10k
Variable
ID: out_dir
Value: /home/gi...r_data_sfcw
GNURadio
THE FREE & OPEN SOFTWARE RADIO ECOSYSTEM
SFCW SAR Engine
F_Start: 1G
F_Stop: 6G
N_Freq: 501
Samp_Rate: 5M
Tx_Gain: 70
Rx_Gain: 55
N_Pos: 64
Dx: 15m
Target_Range: 2.5
N_Settle: 5k
N_Meas: 10k
Out_Dir: /home/g..._data_sfcw
out
QT GUI Frequency Sink
Name: Live SFC...nt position)
FFT Size: 512
Center Frequency (Hz): 0
Bandwidth (Hz): 5M
in
QT GUI Time Sink
Name: SFCW Tone Amplitudes
Number of Points: 1.024k
Sample Rate: 5M
Autoscale: Yes
in

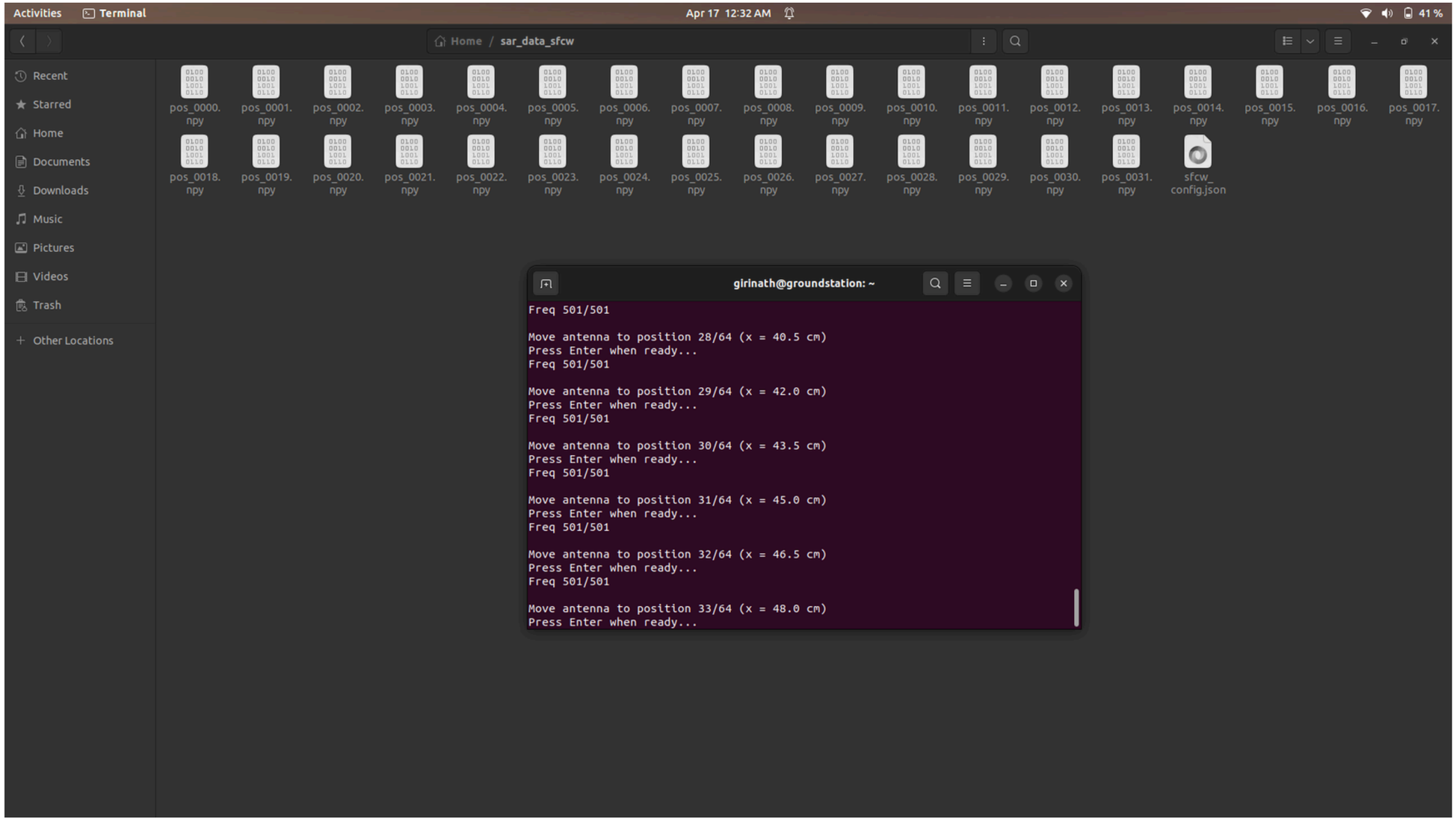
SFCW Working Process

For each antenna position:

1. Sweep frequencies from 1 GHz to 6 GHz
2. At each frequency:
    a. Transmit CW signal
    b. Receive reflected signal
    c. Measure complex response
3. Store response



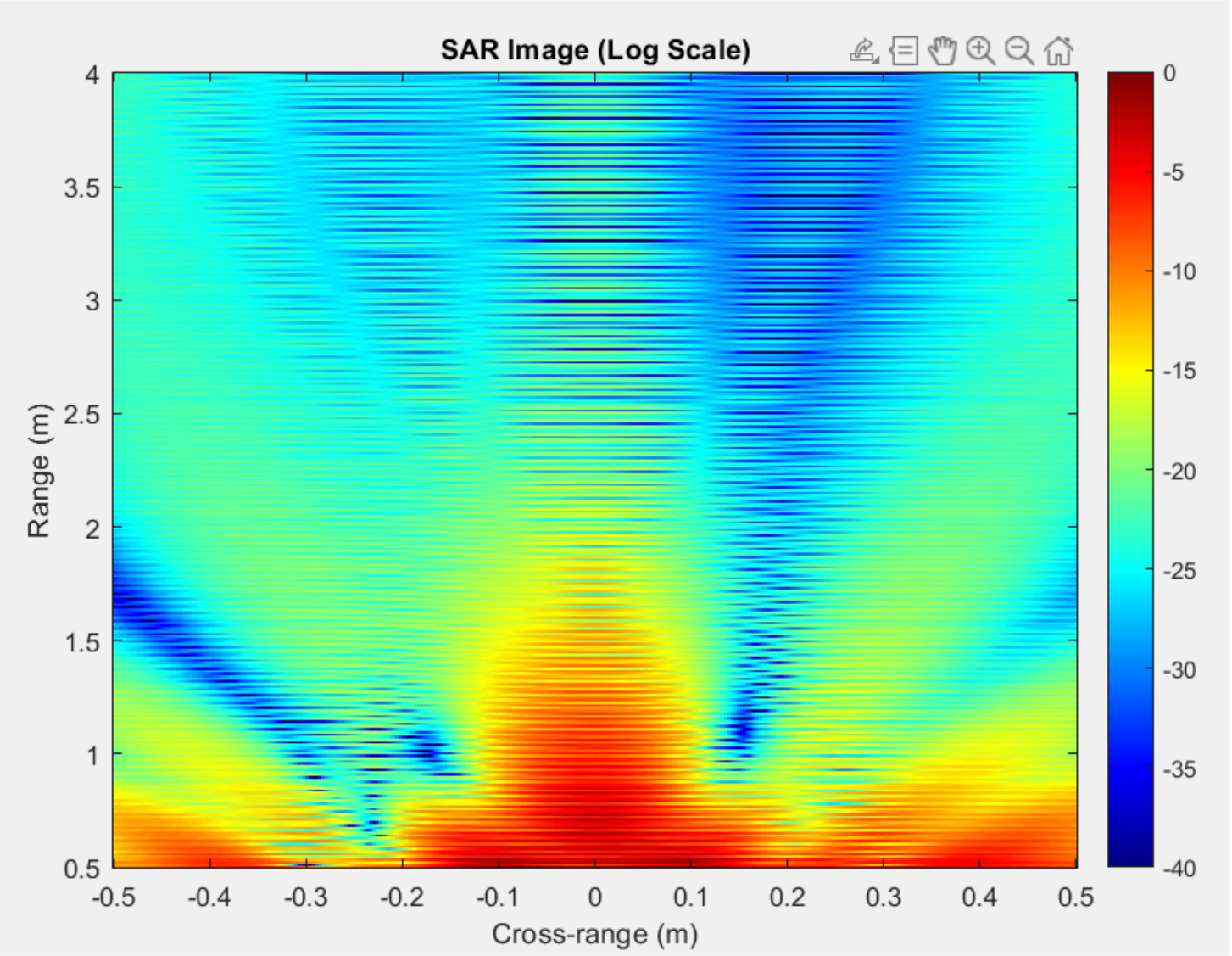
Data Storage Structure



Matrix Structure

	F1	F2	F3	F4	F5	.	.	.	.	F499	F500	F501
X1												
X2												
.												
.												
.												
X64												

RESULTS



1) Load Data

2) Define RADAR Parameters

3) Remove Static Clutter

4) Windowing

5) Imaging Grid

6) Backprojection

7) Normalisation & Display

LITERATURE SURVEY

S No.	Title	Journal	Inference
1	Design and Implemetation of Compact 77 GHz Synthetic Aperture Radar for Drone Based Applications	2019 6th Asia-Pacific Conference on Synthetic Aperture Radar (APSAR)	• The paper demonstrates the design of a **compact 77 GHz millimeter-wave SAR system integrated with a UAV (drone)** to enable lightweight and efficient airborne radar imaging. • Experimental results show that the proposed system can **successfully acquire SAR data and generate radar images while maintaining small size, low weight, and suitability for drone-based remote sensing applications**.
2	FMCW SAR Imaging Based on Software Defined Radio Architecture	2023 International Telecommunications Conference (ITC-Egypt)	• The paper presents an **FMCW Synthetic Aperture Radar (SAR) imaging system implemented using a Software Defined Radio (SDR) architecture**, enabling flexible signal generation, acquisition, and processing. • Experimental results demonstrate that the SDR-based FMCW SAR system can **successfully generate SAR images with lower cost, high flexibility, and easy reconfiguration compared to traditional radar hardware systems**.
3	SAR Image Stitching Method for M×N SAR Sub-images	IEEE 2024	• The paper proposes a SAR image stitching method that combines **phase correlation–based sub-image registration and least-squares global optimization** to accurately align multiple SAR sub-images and determine their global positions. • The approach further applies **gray-scale equalization and seam blending** to remove intensity discontinuities, producing a **large-scene SAR image with improved field of view and higher resolution**.

LITERATURE SURVEY

S No.	Title	Journal	Inference
4	Utilizing Polarization Diversity in GBSAR Data-Based Object Classification	Sensors 2024	• The paper demonstrates the use of a **Ground-Based SAR (GBSAR)** system to classify close-range objects based on their geometric shape and material composition (such as metal, glass, or plastic) by analyzing their unique **dielectric properties and radar cross-sections**. • Experimental results show that by utilizing localized scanning and polarization diversity, the system can successfully **distinguish between targets of the exact same size but different materials**, proving the viability of small-scale, ground-based setups for high-resolution physical classification.
5	Fast-Fourier Time-Domain SAR Reconstruction for Millimeter-Wave FMCW 3-D Imaging	IEEE Transactions on Microwave Theory and Techniques (Volume: 72, Issue: 12, December 2024)	• The paper demonstrates an accurate **phase calibration technique** and a highly efficient time-domain reconstruction algorithm tailored specifically for 3-D **millimeter-wave FMCW SAR imaging**. • Experimental results show that the proposed holographic algorithm can **generate high-resolution 3-D images of targets** and concealed objects at close range. It operates two orders of magnitude faster than existing time-domain methods, making it highly effective for near-field, localized radar setups like your USRP configuration.

REFERENCES

1. Penner, J.F.; Long, D.G. Ground-Based 3D Radar Imaging of Trees Using a 2D Synthetic Aperture. Electronics 2017, 6, 11. https://doi.org/10.3390/electronics6010011
2. G. L. Charvat and L. C. Kempel, "Low-Cost, High Resolution X-Band Laboratory Radar System for Synthetic Aperture Radar Applications," 2006 IEEE International Conference on Electro/Information Technology, East Lansing, MI, USA, 2006, pp. 529-531, doi: 10.1109/EIT.2006.252203. keywords: {Synthetic aperture radar;Laboratories;Radar applications;Radar cross section;Pulse measurements;Costs;Radar measurements;Testing;Instruments;Frequency;FMCW Radar;Synthetic Aperture Radar;SAR;ISAR;Radar Imaging;Low Cost Radar System},
3. M. Lisano et al., "NASA-ISRO Synthetic Aperture Radar (NISAR): Towards Observatory-level Integration and Testing," 2023 IEEE Aerospace Conference, Big Sky, MT, USA, 2023, pp. 1-10, doi: 10.1109/AERO55745.2023.10115885. keywords: {Space vehicles;Observatories;Radar measurements;Spaceborne radar;Instruments;Space missions;NASA}
4. https://in.mathworks.com/help/radar/ug/simulated-land-scenes-for-synthetic-aperture-radar-image-formation.html
5. https://www.gnuradio.org/grcon/grcon19/presentations/USRP_based_X-band_Digital_Beam_Forming_Synthetic_Aperture_Imaging_Radar/Peter%20Stenger%20-%20Synthetic%20Aperture%20Imaging%20Radar.pdf

THANK YOU !